

Erdős Problem #243

EP-243

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $1 \leq a_1 < a_2 < \dots$ be a sequence of integers such that $\lim_{n \rightarrow \infty} \frac{a_n}{a_{n-1}^2} = 1$ and $\sum \frac{1}{a_n} \in \mathbb{Q}$. Then, for all sufficiently large $n \geq 1$, $a_n = a_{n-1}^2 - a_{n-1} + 1$.

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>